

ĐẠI HỌC BÁCH KHOA HÀ NỘI
TRƯỜNG CÔNG NGHỆ THÔNG TIN & TRUYỀN THÔNG



BÁO CÁO MÔN HỌC

Thực hành lập trình hướng đối tượng

IT3103- 750868

BÁO CÁO LAB 004

Phần 2

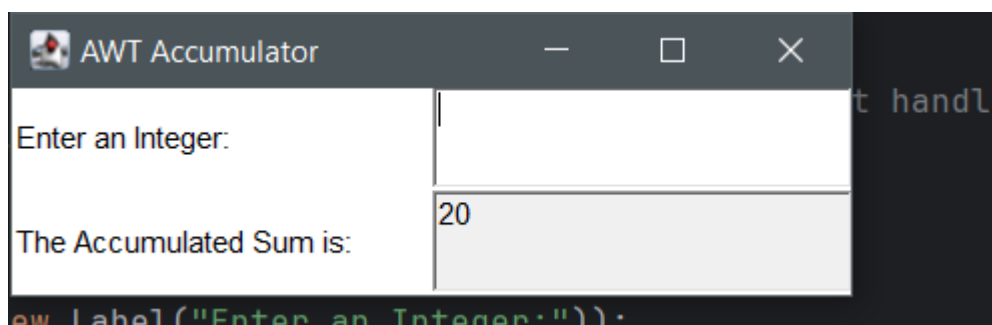
Giảng viên hướng dẫn : Hồ Viết Đức Lương
Sinh viên thực hiện : Hoàng Đức Anh - 20235640

1.1) AWTAccumulator

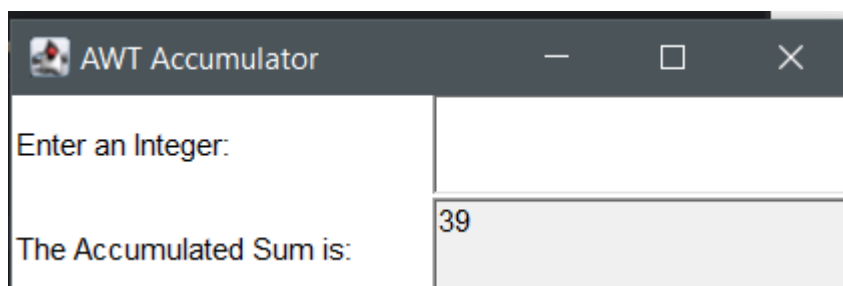
Màn hình khởi tạo :

```
1 package hust.soict.hedspi.swing;
2
3 > import ...
4
5
6
7 public class AWTAccumulator extends Frame {
8     private TextField tfInput; 5 usages
9     private TextField tfOutput; 4 usages
10    private int sum = 0; // Accumulated sum, init to 0 2 usages
11
12    // Constr
13    public AWTAccumulator() {
14        setLabel("Enter an Integer:");
15        setLabel("The Accumulated Sum is:");
16        add(new Label("Enter an Integer:"));
17
18        tfInput = new TextField(10);
19        add(tfInput);
20        tfInput.addActionListener(new TFInputListener());
21
22        add(new Label("The Accumulated Sum is:"));
23    }
24 }
```

Điền : 20

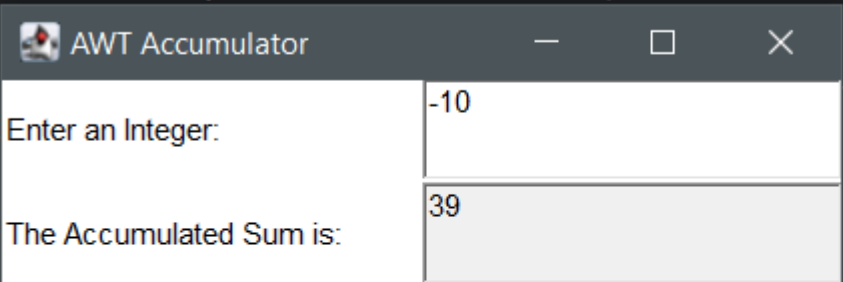


Điền : 19

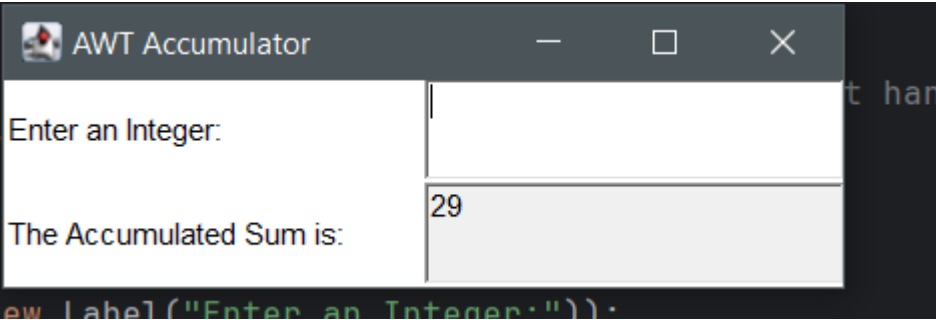


Điền : -10

```
int sum = 0; // Accumulated sum, init to 0 2 usages
```

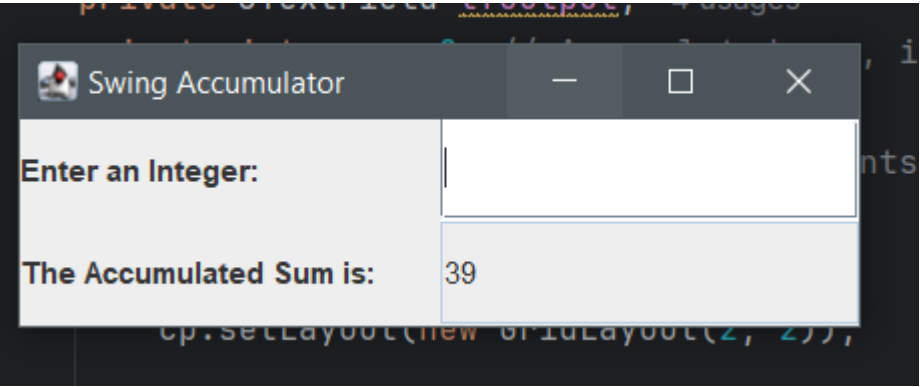


```
(new Label("Enter an Integer:"));  
  
input = new TextField(10);  
(tfInput);  
input.addActionListener(new TFIInputListener());
```

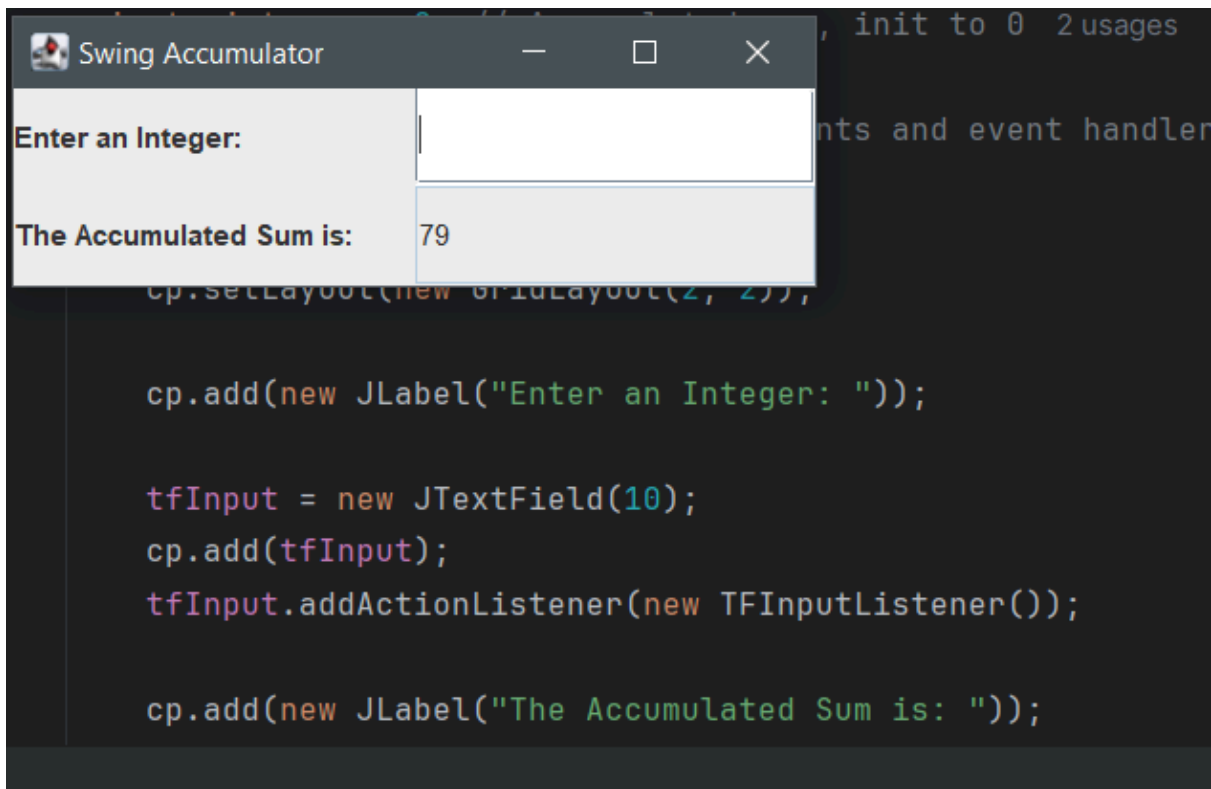


1.2) Swing Accumulator

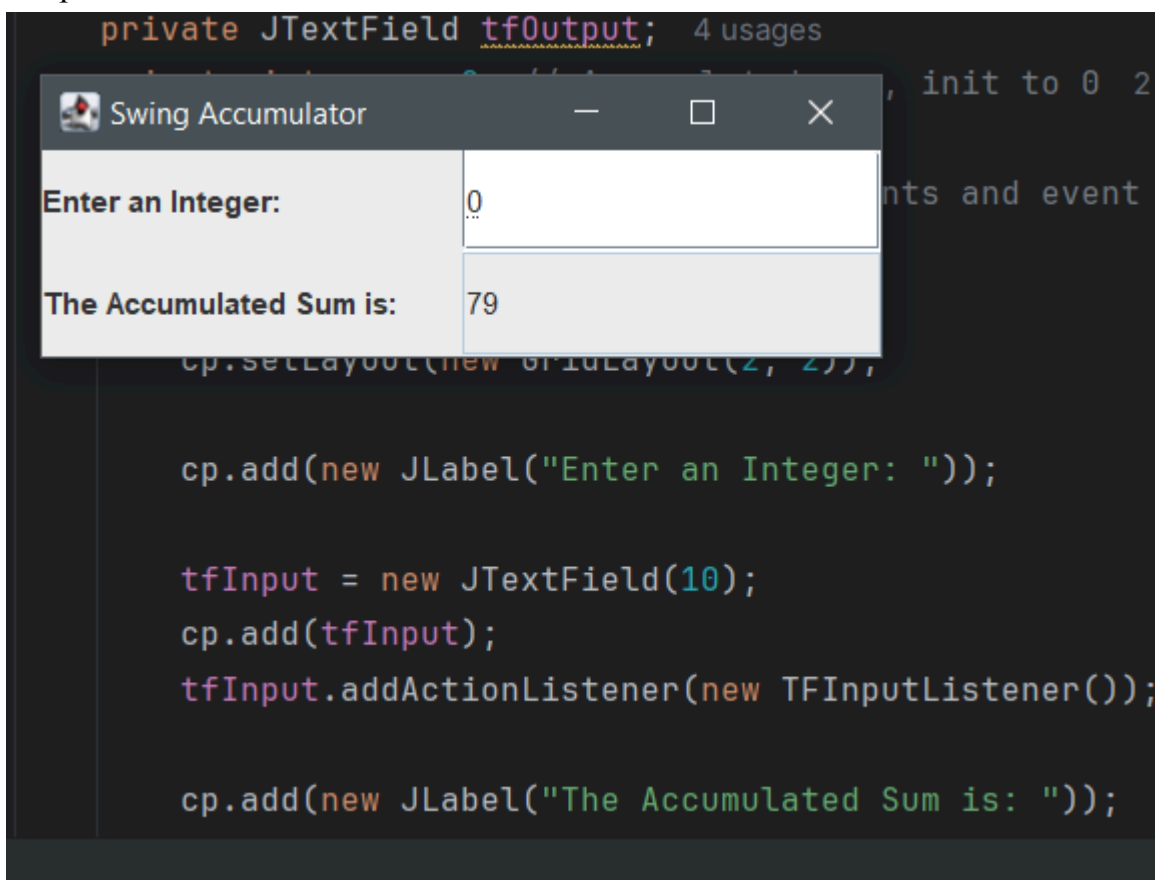
Nhập 39



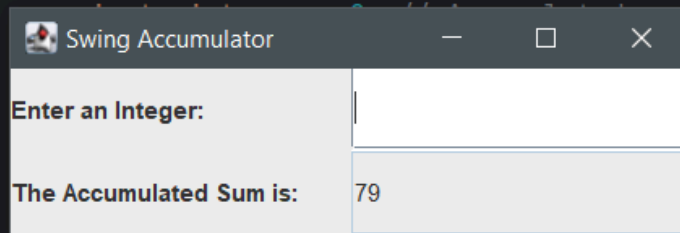
Nhập 40



Nhập 0

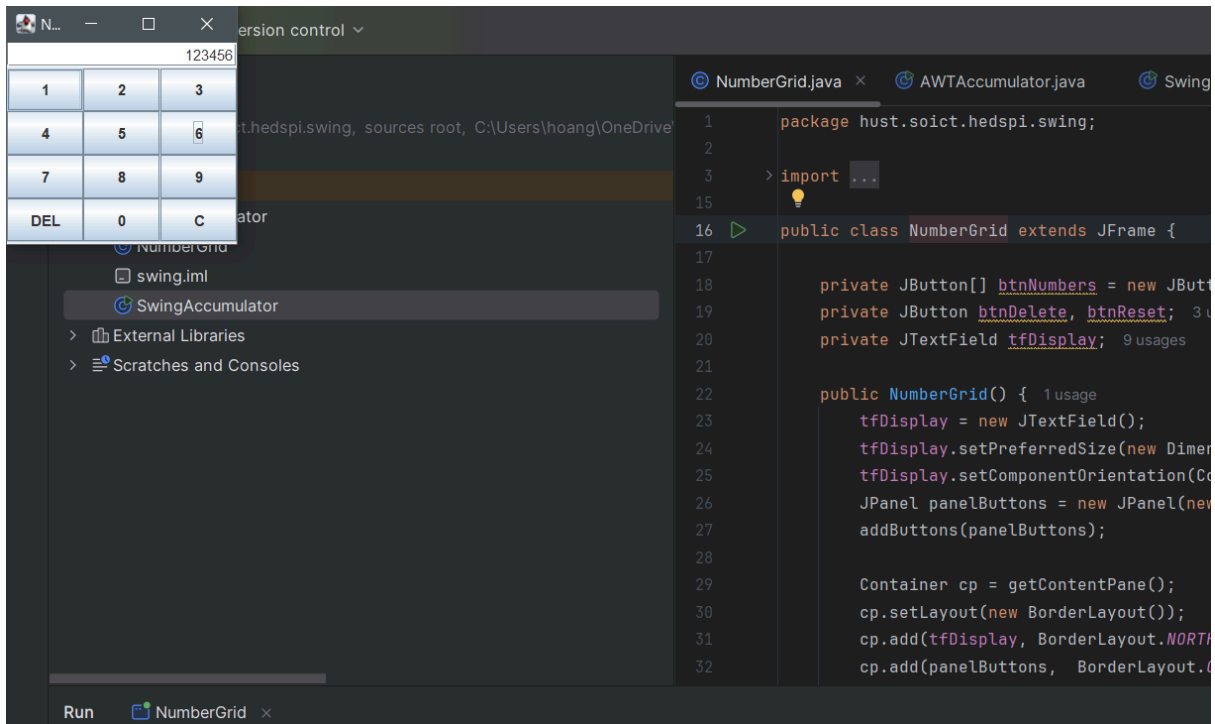


```
1 package hust.soict.hedspi.swing;
2
3 > import ...
7
8 public class SwingAccumulator extends JFrame {
9     private JTextField tfInput; 5 usages
10    private JTextField tfOutput; 4 usages
11    // ... (commented out) //, init to 0 2 usages
12
13    // ... (commented out) //nts and event handlers
14
15    // ... (commented out) //
16    cp.setLayout(new GridLayout(2, 2));
17
18    cp.add(new JLabel("Enter an Integer: "));
19
20    tfInput = new JTextField(10);
21    cp.add(tfInput);
22    tfInput.addActionListener(new TFInputListener());
23
24    cp.add(new JLabel("The Accumulated Sum is: "));
```

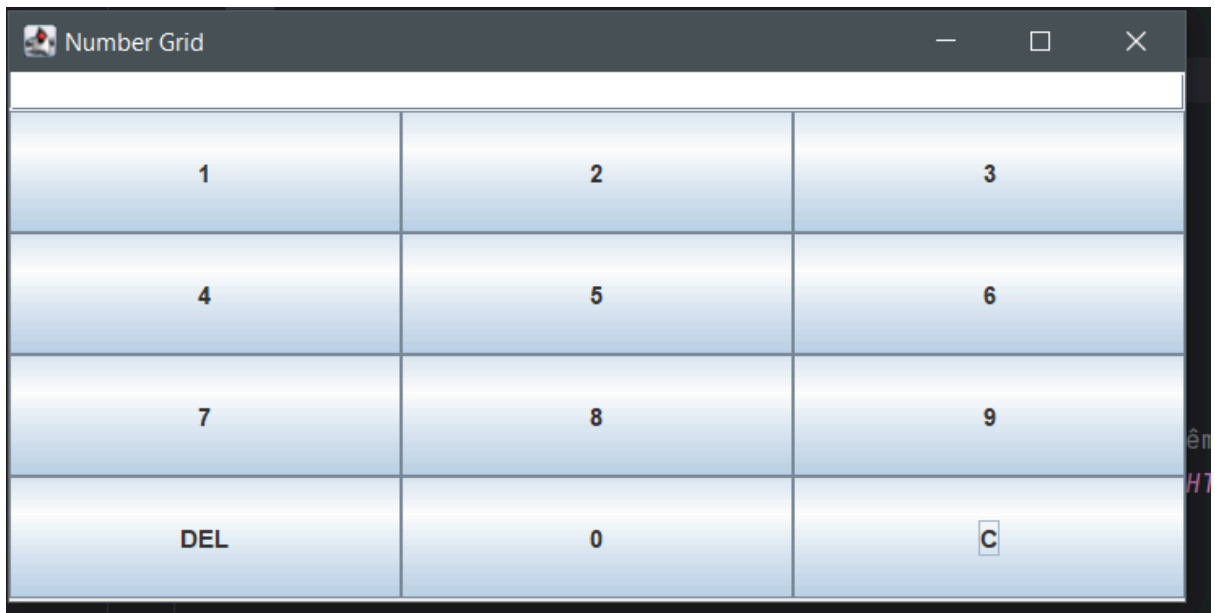


2. NumberGrid:

Nhập : 123456



Bấm C :



Bấm DEL 3 lần :

NumberGrid.java × AWTAccumulator.java SwingAccumulator.java

1 package hust.soict.hedspi.swing;
2
3 > import ...

15 Number Grid

123

1	2	3
4	5	6
7	8	9
DEL	0	C

26
27
28
29 Container cp = getContentPane();
30 cp.setLayout(new BorderLayout());
31 cp.add(tfDisplay, BorderLayout.NORTH);
32 cp.add(panelButtons, BorderLayout.CENTER);